by a comma. The microservice would then see a header with multiple certificate details in it, as in the following example:

```
x-forwarded-client-cert: By=https://gateway.example.org;
[CA]Hash=0d352f0688d3a686e56a72852a217ae461a594ef22e54cb
[CA]551af5ca6d70951bc,By=spiffe://api.natter.local/ns/    #A
[CA]natter-api/sa/natter-api-service;Hash=b26f1f3a5408f7
[CA]61753f3c3136b472f35563e6dc32fefd1ef97d267c43bcfdd1
```

#A The comma separates the two certificate entries

The original client certificate presented to the gateway is the first entry in the header, and the certificate presented by the Istio sidecar proxy is the second. The gateway itself will strip any existing header from incoming requests, so the append behavior is only for internal sidecar proxies. The sidecar proxies also strip the header from new outgoing requests that originate inside the service mesh. These features allow you to use client certificate authentication in Istio without needing to generate or manage your own certificates. Within the service mesh this is entirely managed by Istio, while external clients can be issued with certificates using an external CA.

11.4.5 Mutual TLS with OAuth2

OAuth2 can also support mTLS for client authentication through a new specification (RFC 8705, https://tools.ietf.org/html/draft-ietf-oauth-mtls-17), which also adds support for certificate-bound access tokens,